

Probability Distributions: CDF, PMF, PDF

Math Foundations — Node 23

First, an example

Before the abstract setup, two warm-up distributions you already know.

Bernoulli(0.3) (discrete). A coin landing heads with probability 0.3. Its PMF places mass on two points: $P(X = 1) = 0.3$ and $P(X = 0) = 0.7$. Total mass: $0.3 + 0.7 = 1$, as required.

Uniform(0, 1) (continuous). A point chosen “at random” in $[0, 1]$. Its PDF is the constant $f(x) = 1$ on $[0, 1]$ and 0 elsewhere; the total area is $\int_0^1 1 dx = 1$. Its CDF is the piecewise linear function

$$F_X(x) = 0 \text{ for } x < 0, \quad F_X(x) = x \text{ for } 0 \leq x \leq 1, \quad F_X(x) = 1 \text{ for } x > 1.$$

Read aloud: “F sub X of x equals zero for x less than zero, equals x on the unit interval, and equals one for x greater than one.”

These are the two flavors — discrete (PMF) and absolutely continuous (PDF) — unified by the CDF defined below.

1 Setup

Throughout, $(\Omega, \mathcal{F}, P)$ is a probability space and $X : \Omega \rightarrow \mathbb{R}$ is a real-valued random variable, i.e. a measurable map with respect to the Borel σ -algebra $\mathcal{B}(\mathbb{R})$. The *distribution* (or *law*) of X is the pushforward measure $\mu_X(A) := P(X \in A)$ on $\mathcal{B}(\mathbb{R})$.

2 Cumulative Distribution Function

Definition 1 (CDF). *The cumulative distribution function of X is*

$$F_X(x) := P(X \leq x) = \mu_X((-\infty, x]), \quad x \in \mathbb{R}.$$

Read aloud: “F sub X of x is defined as the probability that X is at most x, which equals mu sub X of the interval from minus infinity to x, for every real x.”

Theorem 1 (Properties of the CDF). *The CDF F_X of any real-valued random variable X satisfies:*

- (i) **Non-decreasing:** $x \leq y \implies F_X(x) \leq F_X(y)$.
- (ii) **Right-continuous:** $\lim_{y \downarrow x} F_X(y) = F_X(x)$ for every $x \in \mathbb{R}$.
- (iii) **Limits:** $\lim_{x \rightarrow -\infty} F_X(x) = 0$ and $\lim_{x \rightarrow +\infty} F_X(x) = 1$.

Proof. (i) Monotonicity. If $x \leq y$, then $\{X \leq x\} \subseteq \{X \leq y\}$, so by monotonicity of the probability measure,

$$F_X(x) = P(X \leq x) \leq P(X \leq y) = F_X(y).$$

Read aloud: “F sub X of x equals the probability that X is at most x, which is at most the probability that X is at most y, which equals F sub X of y.”

(ii) Right-continuity. Fix $x \in \mathbb{R}$ and let $(y_n)_{n \geq 1}$ be any sequence with $y_n \downarrow x$. Define $A_n := \{X \leq y_n\}$. Since y_n is decreasing, $A_1 \supseteq A_2 \supseteq \dots$, and

$$\bigcap_{n=1}^{\infty} A_n = \bigcap_{n=1}^{\infty} \{X \leq y_n\} = \{X \leq x\},$$

Read aloud: “the intersection over n from one to infinity of A sub n equals the intersection of the events X is at most y sub n, which equals the event X is at most x.” because $y_n \rightarrow x$ from above implies $X(\omega) \leq y_n$ for all n iff $X(\omega) \leq \inf_n y_n = x$. The sets A_n have finite measure (bounded by 1), so continuity from above of P (a consequence of countable additivity) gives

$$\lim_{n \rightarrow \infty} P(A_n) = P\left(\bigcap_{n=1}^{\infty} A_n\right) = P(X \leq x) = F_X(x).$$

Read aloud: “the limit as n tends to infinity of P of A sub n equals P of the intersection of all A sub n, which equals the probability X is at most x, which equals F sub X of x.” Hence $\lim_{n \rightarrow \infty} F_X(y_n) = F_X(x)$. Since this holds for every such sequence, F_X is right-continuous at x .

(iii) Limits at $\pm\infty$. Take any sequence $x_n \uparrow +\infty$ and let $B_n := \{X \leq x_n\}$. Then $B_1 \subseteq B_2 \subseteq \dots$ and

$$\bigcup_{n=1}^{\infty} B_n = \{X < +\infty\} = \Omega,$$

Read aloud: “the union over n from one to infinity of B sub n equals the event X is finite, which equals the whole sample space Omega.” since X is real-valued. By continuity from below of P (again, countable additivity),

$$\lim_{n \rightarrow \infty} F_X(x_n) = \lim_{n \rightarrow \infty} P(B_n) = P(\Omega) = 1.$$

Read aloud: “the limit as n tends to infinity of F sub X at x sub n equals the limit of P of B sub n, which equals P of Omega, which equals one.” Similarly, take $x_n \downarrow -\infty$ and $C_n := \{X \leq x_n\}$. Then $C_1 \supseteq C_2 \supseteq \dots$ and

$$\bigcap_{n=1}^{\infty} C_n = \{X = -\infty\} = \emptyset,$$

Read aloud: “the intersection over n from one to infinity of C sub n equals the event X equals minus infinity, which is the empty set.” so by continuity from above,

$$\lim_{n \rightarrow \infty} F_X(x_n) = P(\emptyset) = 0.$$

Read aloud: “the limit as n tends to infinity of F sub X at x sub n equals P of the empty set, which equals zero.”

This proves all three properties. □

Remark 1. *The converse also holds: every function $F : \mathbb{R} \rightarrow [0, 1]$ satisfying (i)–(iii) is the CDF of some random variable. (Construction: take $\Omega = (0, 1)$ with Lebesgue measure and define $X(u) = \inf\{x : F(x) \geq u\}$, the generalized inverse. This is the inverse-CDF sampling method.)*

3 Probability Mass Function (Discrete Case)

Definition 2 (PMF). A random variable X is discrete if there exists a countable set $S \subset \mathbb{R}$ with $P(X \in S) = 1$. Its probability mass function is

$$p_X(k) := P(X = k), \quad k \in S.$$

Read aloud: “ p sub X of k is defined as the probability that X equals k , for every k in the support set S .”

It satisfies $p_X(k) \geq 0$ and $\sum_{k \in S} p_X(k) = 1$, and the CDF is the running sum

$$F_X(x) = \sum_{k \in S, k \leq x} p_X(k).$$

Read aloud: “ F sub X of x equals the sum, over all k in S with k at most x , of p sub X of k .” The CDF of a discrete RV is a step function with jumps of size $p_X(k)$ at each $k \in S$.

4 Probability Density Function (Continuous Case)

Definition 3 (PDF). A random variable X is absolutely continuous if there exists a measurable function $f_X : \mathbb{R} \rightarrow [0, \infty)$ such that

$$F_X(x) = \int_{-\infty}^x f_X(t) dt \quad \text{for all } x \in \mathbb{R}.$$

Read aloud: “ F sub X of x equals the integral from minus infinity to x of f sub X of t d t , for every real x .” Such an f_X is called a probability density function of X and satisfies $\int_{\mathbb{R}} f_X = 1$. It is unique up to a Lebesgue-null set, and where F_X is differentiable, $f_X(x) = F'_X(x)$.

For absolutely continuous X and any $a < b$,

$$P(a < X \leq b) = F_X(b) - F_X(a) = \int_a^b f_X(t) dt.$$

Read aloud: “the probability that X lies strictly above a and at most b equals F sub X of b minus F sub X of a , which equals the integral from a to b of f sub X of t d t .” In particular $P(X = x) = 0$ for every single point x , so $f_X(x)$ is a *density*, not a probability — it can exceed 1 (e.g. $\text{Uniform}(0, 0.5)$ has $f_X = 2$ on $[0, 0.5]$).